



Tax multipliers: Pitfalls in measurement and identification



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ABSTRACT

To analyze the pitfalls in measuring and identifying tax shocks, we build a novel value-added tax rate dataset for the period 1980–2009. The problem of identification (i.e., changes in tax policy not triggered by output fluctuations) is clearly disentangled from the problem of measurement (i.e., finding a tax policy variable under the direct control of policymakers). On the identification front, our results favor the use of narratives à la [Romer and Romer \(2010\)](#). On the measurement front, our findings support the use of tax rates as the true measure of tax policy as opposed to revenue-based measures, such as cyclically adjusted revenues.

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1. Introduction

In the aftermath of the global financial crisis and ensuing recession triggered by the fall of Lehman Brothers on September 15, 2008, many governments across the world implemented aggressive countercyclical fiscal policies. More recently, large fiscal deficits and concerns about debt sustainability in industrial countries, particularly in Europe, have shifted the tone of the discussion from stimulus to fiscal adjustment. Since then – and following the influential paper by [Giavazzi and Pagano \(1990\)](#) – there has also been a revival of studies suggesting the possibility that fiscal adjustments might have no effect on output, or be even expansionary, especially when driven by spending cuts.¹ As a result of the policy focus first on fiscal stimulus and later on fiscal consolidation, there has also been a flourishing of studies estimating fiscal multipliers, both on the spending and the taxation side.

The main challenge and point of contention among researchers has been how to address the possible endogeneity of fiscal policy or, to put it differently, how to identify fiscal policy shocks (i.e., changes in fiscal policy variables that are not directly or indirectly related to output changes). Two main approaches have been used to address these endogeneity problems. The first approach has followed the seminal contribution of [Blanchard and Perotti \(2002\)](#) by imposing short term restrictions in the context of structural vector autoregressions (SVAR). On the expenditure side, the Blanchard–Perotti approach assumes that discretionary government spending requires at least one quarter to respond to news about the state of the economy. On the taxation side, the output elasticity of tax revenues is used to differentiate “discretionary” changes in

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¹ See, for example, [McDermott and Wescott \(1996\)](#), [Giavazzi et al. \(2000\)](#), [von Hagen and Strauch \(2001\)](#), [von Hagen et al. \(2002\)](#), [Lambertini and Tavares \(2005\)](#), [Ardagna \(2004\)](#), [Alesina and Ardagna \(2010\)](#), and [IMF \(2010\)](#).

taxation (also referred to as changes in cyclically adjusted revenues) from those driven by endogenous reactions of tax revenues to output fluctuations. The Blanchard–Perotti approach is by far the most popular identification strategy due to its ease of implementation and data availability (e.g., Perotti, 2004; Favero and Giavazzi, 2007, 2012; Mountford and Uhlig, 2009; Ilzetzki, 2011; Ilzetzki et al., 2013). This strategy has been criticized on the basis that most changes in government spending and taxes are actually anticipated by agents (e.g., Ramey and Shapiro, 1998; Leeper et al., 2008; Ramey, 2011; Auerbach and Gorodnichenko, 2013; Riera-Crichton et al., 2015).

The second approach has developed around the so-called “natural experiment” approach. This identification strategy assumes that some exogenous changes in fiscal policy are, in fact, observable. On the spending side, studies have typically focused on particular spending categories, such as military buildups, that are unlikely to respond to output fluctuations (e.g., Barro, 1981; Ramey and Shapiro, 1998; Hall, 2009; Ramey, 2011; Barro and Redlick, 2011). On the taxation side, Romer and Romer (2010), hereafter RR, use the narrative record – such as presidential speeches and congressional reports – to identify, on a case-by-case basis, the nature of legislated federal tax changes in the United States from 1945 to 2007. RR identify tax changes exogenous to the business cycle, because they either were passed for long-run growth reasons or involve increases seeking to reduce an inherited budget deficit. On tax changes induced by long-run growth considerations, they argue that: “[t]he quintessential exogenous change might be a tax cut motivated by a belief that lower marginal tax rates will raise output in the long run. Such an action is fundamentally different from the countercyclical actions [...] because the goal is to raise normal growth, not to offset shocks acting to reduce growth relative to normal.” Regarding deficit-driven tax changes, they argue that “[a]n inherited deficit reflects past economic conditions and budgetary decisions, not current conditions or spending changes. If policymakers raise taxes to reduce such a deficit, this is not a change motivated by a desire to return growth to normal or to prevent abnormal growth. So it is exogenous.”

Our analysis contributes to the literature on tax multipliers. Building on the above discussion on *identification*, the paper revisits the merits of the SVAR vs. the “natural experiment” approach.² To implement the latter, we take as the starting point the action-based fiscal consolidation episodes identified by an IMF study in 15 industrial countries for the period 1980–2009, and then develop our own narrative to ensure exogeneity. This strategy is in the spirit of RR’s identification of tax changes driven by inherited fiscal deficits.

The main and novel contribution of this paper, however, is related to the much less explored issue of *measurement* of tax policy (i.e., finding a tax policy variable under the direct control of the policymaker). As discussed in Kaminsky et al. (2004) and Vegh and Vuletin (2015), an obvious, yet critical, observation is that policymakers’ main tax instrument is the tax rate. While policymakers control and legislate on tax rates, tax revenues are a policy *outcome* not under the policymakers’ direct control. To fix ideas, define real tax revenues (R) as follows:

$$R_t = \text{TAX RATE}_t \cdot \text{TAX BASE}_t, \quad (1)$$

where TAX BASE_t denotes the real tax base. The policymaker controls TAX RATE but not TAX BASE and, hence, does not control R .³ Taking logarithmic changes, it follows from Eq. (1) that

$$r_t - r_{t-1} = (\text{tax rate}_t - \text{tax rate}_{t-1}) + (\text{tax base}_t - \text{tax base}_{t-1}), \quad (2)$$

where r , tax rate , and tax base are the log of real tax revenues (R), tax rate (TAX RATE), and the real tax base (TAX BASE), respectively. Eq. (2) indicates that the percentage change in tax revenues ($\Delta r_t \equiv r_t - r_{t-1}$) can be decomposed into the sum of the percentage change in the tax rate ($\Delta \text{tax rate}_t \equiv \text{tax rate}_t - \text{tax rate}_{t-1}$) and the percentage change in the tax base ($\Delta \text{tax base}_t \equiv \text{tax base}_t - \text{tax base}_{t-1}$).

Given the lack of readily available cross country data on tax rates, the standard measure utilized in the tax multiplier literature to capture discretionary changes in tax policy is the so-called cyclically adjusted revenues (e.g., Giavazzi and Pagano, 1990; Alesina and Perotti, 1997; Blanchard and Perotti, 2002; Alesina and Ardagna, 2010; Favero and Giavazzi, 2012; Perotti, 2012; Ilzetzki, 2011). The change in cyclically adjusted revenues is typically calculated as:

$$\Delta \text{cyclically-adjusted}_t = r_t - r_{t-1} - \eta(y_t - y_{t-1}), \quad (3)$$

where y is the log of real output and η is the historical (i.e., average) output elasticity of tax revenues. The first two terms on the right-hand side of (3) denote the percentage change in tax revenues. The third term aims at capturing the percentage change in tax revenues associated with GDP-driven changes in the tax base. In principle, this cyclically adjusted measure offers an intuitive way of dealing with the fact that the tax base moves endogenously with the business cycle. The idea is, of course, that once tax revenues are cyclically adjusted, changes will reflect the discretionary action of policymakers. Indeed, assuming that $\text{tax base}_t = \eta \cdot y_t$, it follows that $\Delta \text{cyclically-adjusted}_t = \Delta \text{tax rate}_t$.^{4,5} In other words, cyclically adjusted revenue changes seem to capture discretionary changes in tax policy (i.e. changes in tax rates).

² In this dimension, our paper is thus related to the recent contribution of Mertens and Ravn (2014), which tries to reconcile SVAR and narrative estimates of tax multipliers. The authors use narrative measures as proxies for structural shocks and trace differences in the magnitude of tax multipliers to incorrect assumptions on the output elasticity of tax revenues and measurement errors in narrative series of tax shocks.

³ This concern is particular to tax policy, since government spending is, in principle, the appropriate policy instrument on the spending side.

⁴ Assuming that $\text{tax base}_t = \eta \cdot y_t$, implies, of course, that η is the elasticity of the tax base with respect to GDP.

⁵ To see this, replace (2) in (3) and use the fact that $\Delta \text{tax base}_t = \eta \cdot \Delta y_t$.

Table 1

Identification of exogenous fiscal shocks vs. measurement of tax policy.

		Measurement of tax policy (Finding a tax policy variable under the direct control of the policymaker)	
		Tax rate	Cyclically-adjusted revenue
Identification of exogenous fiscal shocks (Fiscal policy changes that are not the result of policymakers responding to output fluctuations)	"Natural experiment" (à la Romer and Romer, 2010)	• No endogeneity. • No measurement error.	• No endogeneity. • Possible measurement error, especially due to changes in non-policy factors.
	Structural VAR (à la Blanchard and Perotti, 2002)	• Endogeneity problems if tax changes are forward-looking/anticipated. • No measurement error.	• Endogeneity problems if tax changes are forward-looking/anticipated. • Possible measurement error, especially due to changes in non-policy factors.

While appealing at first, revenue-based measures of tax policy, such as cyclically adjusted revenues, suffer from important measurement errors. Mirroring the discussion in the growth literature on the Solow residual, the cyclically adjusted measure implicitly attributes any change in revenues not associated with the estimated change in the tax base to policymakers' discretionary behavior. This source of measurement error would include, among many others, structural breaks, non-linearities and/or changes in agents' behavior over the business cycle, changes in η over time, changes in taxpayers' willingness/possibility to evade taxes, and changes in income distribution.

In sum, there is no good substitute for obtaining data on tax rates themselves when it comes to measuring changes in the tax policy instrument. To this effect – and to assess the practical relevance of this measurement issue – we build a novel tax rate dataset and the corresponding cyclically adjusted revenue measure and compare the implications in terms of the size of the tax multiplier. Given the lack of readily available data on average marginal individual and/or corporate income tax rates for this group of countries, the main focus is on building a new quarterly VAT rate series. The sample includes 14 industrial countries for the period 1980–2009.⁶

Table 1 summarizes our analysis regarding the *identification* of exogenous fiscal policy shocks (i.e., fiscal policy changes that are not the result of policymakers responding to output fluctuations) as separate from problems related to the *measurement* of tax policy (i.e., finding a tax policy variable under the direct control of the policymaker).

The paper's main results may be summarized as follows. First, on the *identification* of tax shocks (i.e., "natural experiment" vs. SVAR), our findings clearly suggest that the "natural experiment" approach is the more accurate strategy to identify exogenous fiscal shocks. As in RR, when using SVAR as our identification strategy, various tax series respond significantly to output fluctuations. These results support previous critiques of the Blanchard–Perotti identification, suggesting that the, presumably, unanticipated changes in fiscal policy (at a one quarter horizon) may actually be anticipated by agents. In contrast, when using exogenous fiscal consolidation episodes as the identification strategy, tax series do not respond to output fluctuations. These findings validate the "natural experiment" approach since tax changes do not seem to be driven by policymakers' short-run reactions to GDP fluctuations.

Second, on the *measurement* of tax policy (i.e., cyclically adjusted revenues vs. tax rates), our findings strongly support the use of changes in tax rates as a measure of tax policy. Indeed, in our analysis, cyclically adjusted revenues consistently fail to proxy for changes in tax rates. In fact, in more than 70% of the countries, cyclically adjusted revenue changes are negatively autocorrelated at the quarterly frequency. This would indicate (under the common presumption that this measure proxies for changes in discretionary tax policy) that policymakers' discretionary policy changes are reversed every quarter! Not surprisingly, these measurement differences have important implications for tax multipliers. Multipliers based on tax rate increases are always negative (i.e., tax policy is contractionary). In sharp contrast, tax multipliers based on cyclically adjusted revenues are close to zero or even positive.

Third, most recent tax studies (e.g., Favero and Giavazzi, 2012; Caldara, 2011; Ilzetzki, 2011) rely on SVAR and cyclically adjusted revenue changes (bottom-right cell in Table 1). In line with most of these papers, when using this identification strategy tax multipliers are found to be not negative. In fact, a one unit increase in tax revenues raises output by 0.19 on impact and 0.80 after four quarters. When focusing on exogenous fiscal consolidation and cyclically adjusted revenue

⁶ The United States is excluded from the IMF's 15 country sample because it does not have federal VAT.

changes (top-right cell in Table 1), tax multipliers also tend to be positive (i.e., tax policy is expansionary). When using SVAR and changes in tax rates (bottom-left cell in Table 1), tax multipliers are negative (i.e., tax policy is contractionary). The multiplier is -0.31 on impact and -1.17 after three quarters. When using exogenous fiscal consolidation and tax rate changes (top-left cell in Table 1), tax multipliers are strongly negative. A one unit revenue increase shock reduces output by 0.66 on impact and 3.56 after three quarters. This wide range of results shows the importance of the strategy used to identify exogenous fiscal shocks and, more importantly for our purposes, the measurement of tax policy.

The rest of the paper is organized as follows. Section 2 discusses data and identification. Section 3 computes tax multipliers using exogenous fiscal consolidation episodes (to identify exogenous fiscal shocks) and tax rates (to measure tax policy) and performs numerous robustness tests. Section 4 examines the implications of alternative empirical strategies regarding the identification of tax shocks and measurement of tax policy. Section 5 concludes.

2. Data and identification

After introducing the dataset on VAT taxation, this section discusses the main identification strategy and then presents the cyclically adjusted data and compares it to our data on tax rate changes.⁷

2.1. VAT data

The VAT dataset covers 14 industrial countries at a quarterly frequency for the period 1980–2009.⁸ Overall – and as depicted in Fig. 1 – there are 42 changes in the VAT rate (32 increases and 10 reductions). Excluding zeros, the median tax rate change is 1 percentage point and the average is 0.7 percentage points. The largest tax rate reduction and tax rate increase are 3.5 and 3 percentage points, respectively. For the same group of countries (though sample periods differ depending on data availability), we also obtained data for VAT revenues (for the purposes, as discussed below, of computing cyclically adjusted revenues).

2.2. Identification of exogenous fiscal consolidation VAT rate changes

Our “narrative-based” empirical strategy relies on the identification of fiscal consolidation episodes carried out in the October 2010 *World Economic Outlook* by the IMF, which covers the period 1980–2009. Examining a variety of documents (including OECD Economic Surveys, IMF Staff Reports, IMF Recent Economic Developments reports, country budget documents, and additional country-specific sources), this IMF study characterizes a fiscal consolidation episode as a situation “in which the government implemented tax hikes, [...] or spending cuts [...] to reduce the budget deficit and put public finances on a more sustainable footing” (IMF, 2010, p. 96).

As a first approximation, it would seem natural to view changes in the VAT rate associated with the fiscal consolidation episodes identified by the IMF (2010) as exogenous to the business cycle (i.e., to quarterly changes in GDP in our case) since such tax changes are presumably part of a fiscal package primarily designed to deal with chronic or inherited budget deficits. There are two scenarios, however, in which this exogeneity assumption may not be valid (in which case the underlying identification strategy would not be correct). First, there may be potential links between the *current* fiscal balance and tax policy. Specifically, it seems plausible that a sudden (i.e., within the quarter) and large change in the fiscal balance may trigger a change in the VAT rate in the same quarter. This could take the form of a sudden deterioration in the fiscal balance caused by a fall in economic activity that induces an increase in the VAT rate to close the fiscal gap, or a sudden improvement in the fiscal accounts that triggers a reduction in tax rates for political economy considerations. Second, an increase in the VAT rate aimed at offsetting other factor(s) that may move output growth away from normal. These factors may include (i) a cut in other tax rates (e.g., a reduction in the employers' contributions to social security) and/or (ii) additional public spending (e.g., job creation programs).

In light of these two possible scenarios – and following the narrative approach – we further analyze the nature of the 19 VAT rate changes that occurred in the episodes of fiscal consolidation identified by the IMF (hereafter “fiscal consolidation tax rate changes”).⁹ Bearing in mind the first concern raised above, when the narrative suggests that changes in VAT rates were triggered by a sudden fiscal deterioration (improvement) resulting from a downturn (upturn) in economic activity, they are classified as endogenous. Regarding the second concern, when the narrative suggests that increases in VAT rates were intended to offset the expansionary effects of government spending or other tax cuts (not included in our set of

⁷ For details on other (standard) data used in this paper, see Online Appendix A.1.

⁸ See Online Appendix A.1 for data sources and sample periods. The countries included are Australia, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, Portugal, Spain, Sweden, and United Kingdom. Other long-standing OECD countries were not included in the analysis because they are not part of the IMF (2010) study that identifies fiscal consolidation episodes. This IMF study covers 15 countries (the 14 countries listed above and the United States) for the period 1980–2009. The United States was not included in our sample because it does not have a federal VAT. The VAT rate for Canada corresponds to the federal VAT rate.

⁹ More precisely, we match our sample of VAT rate changes to the IMF's sample of fiscal consolidation episodes, and look at the intersection of the two (i.e., fiscal consolidation episodes in which the VAT rate changed).

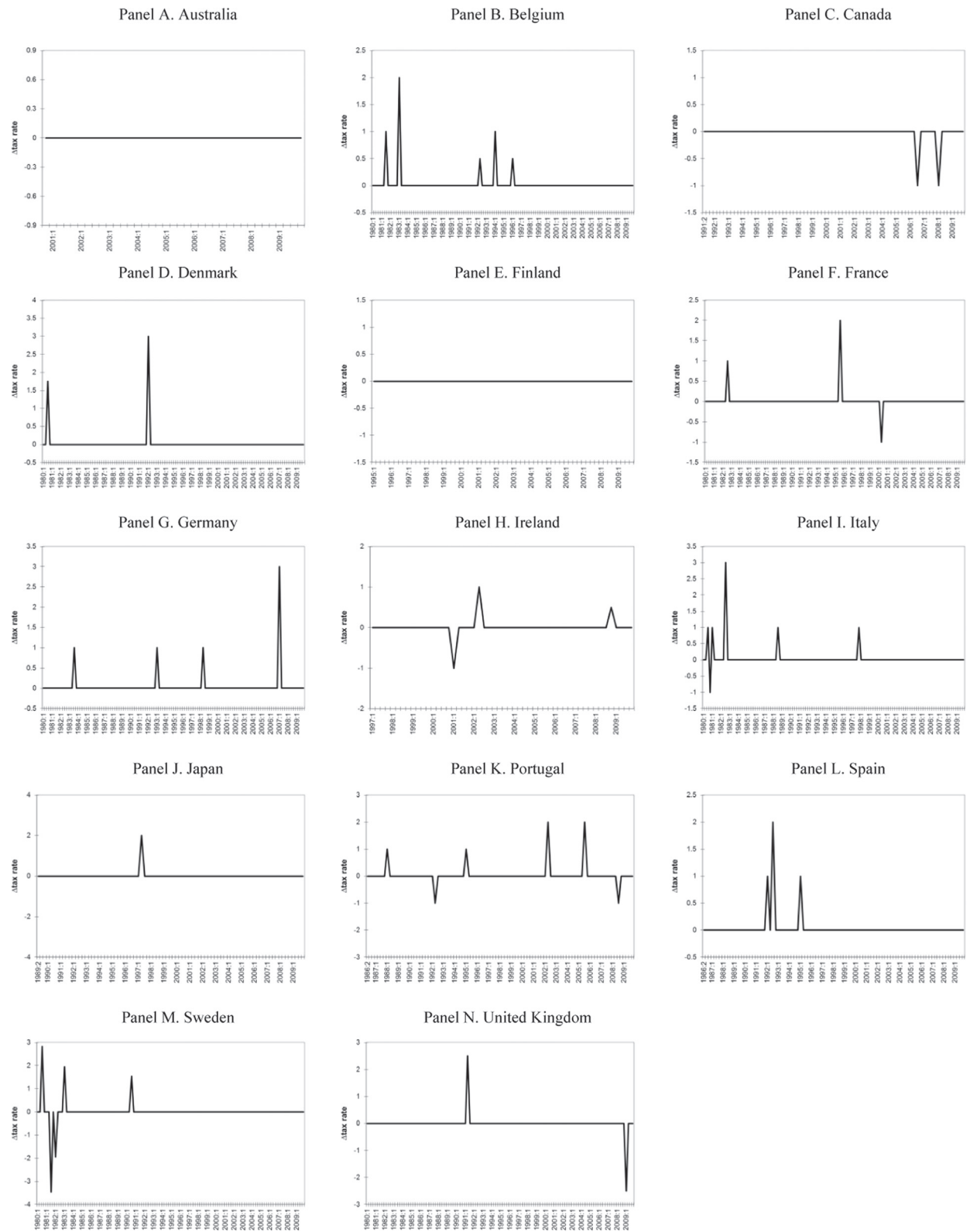


Fig. 1. Tax rate changes.

Table 2
Tax narratives.

Country (1)	Date of implementation of VAT change (2)	Δ VAT rate (in percentage-points) (3)	Brief narrative (4)	Date of passage of VAT change (for exogenous cases only) (5)
Belgium	January 1, 1983	2 (17–19%)	Exogenous. Fiscal deficit rose from 12% of GDP in 1980 to 16% of GDP in 1981. Public debt reached 127% of GDP in 1982.	October 10, 1983
Belgium	April 1, 1992	0.5 (19–19.5%)	Exogenous. Belgium's general government deficit peaked at over 16% of GNP in 1981. As a result, fiscal policy for the rest of the decade first focused on stabilizing and then reducing the public debt ratio.	March 1, 1992
Belgium	January 1, 1994	1 (19.5–20.5%)	Endogenous. A reduction in employers' contributions to social security to restore competitiveness and stimulate employment was funded by a VAT hike.	
Belgium	January 1, 1996	0.5 (20.5–21%)	Exogenous. The VAT increase was part of a medium-term fiscal plan to reduce the deficit to the Maastricht limit of 3% of GDP.	October 1, 1995
France	August 1, 1995	2 (18.6–20.6%)	Endogenous. The VAT increase was put in place partly to pay for a job creation program and was offset by other changes "seen as emergency measures to combat a "calamitous" economic situation". (Dejevsky, 1995).	
France	April 1, 2000	–1 (20.6–19.6%)	Endogenous. The VAT reduction was motivated by pressure to lower taxes in the face of fiscal surpluses in the context of a booming economy.	
Germany	July 1, 1983	1 (11–12%)	Endogenous. Fiscal policy in 1983 was a combination of consolidation efforts and measures to stimulate the economy.	
Germany	January 1, 1993	1 (14–15%)	Exogenous. The 1993 VAT increase was intended to address the deficit resulting from the 1990 reunification.	February 1, 1992
Germany	April 1, 1998	1 (15–16%)	Exogenous. Increasing social expenditure and declining tax revenue exacerbated the large budget deficits run following reunification (which were above the Maastricht treaty). The VAT was increased to address this situation.	
Germany	January 1, 2007	3 (16–19%)	Exogenous. High-cost entitlement programs contributed to rising debt-to-GDP ratio, projected at 67.5% in 2006. The increase in VAT was meant to improve the fiscal situation.	June 16, 2006
Ireland	December 1, 2008	0.5 (21–21.5%)	Exogenous. "Well before the crisis hit, public finances had developed serious structural weaknesses" (IMF, 2009). With the fiscal deficit projected to reach 15% of GDP in 2008, the government increased taxes and cut spending.	October 14, 2008
Italy	October 1, 1997	1 (19–20%)	Exogenous. To ensure entry into the EMU – and with the fiscal deficit for 1997 projected at 6.7% of GDP and the debt-ratio still at 122% of GDP – the Italian government decided to increase the pace of fiscal consolidation with additional revenue measures such as an increase in VAT.	Sept. 1, 1997
Japan	April 1, 1997	2 (3–5%)	Exogenous. As a result of fiscal stimulus in the early 1990s, the government deficit had increased to 4.2% of GDP by 1996. In 1997, policy shifted toward deficit reduction and returning to a stable fiscal position, which included an increase in the VAT.	June 25, 1996
Portugal	June 5, 2002	2 (17–19%)	Exogenous. Portugal's Stability Program carried a commitment to a balanced structural budget by 2004. The high and chronic fiscal deficits since the early 1990s forced the government to take several measures to correct the deficit in 2002.	May 5, 2002
Portugal	July 1, 2005	2 (19–21%)	Exogenous. "The government, reluctant to cut other spending, defended the increase in the VAT rate from 19–21% by arguing it had to raise taxes due to European Community pressure for quick improvement on the deficit front" (Frasquilho, 2005).	June 1, 2005
Spain	January 1, 1992	1 (12–13%)	Exogenous. In order to adjust fiscal deficits to Maastricht Treaty levels, the Spanish authorities reacted to the fiscal deterioration of 1990 and 1991 by reducing unemployment compensation benefits and increasing the VAT on two occasions in 1992.	December 30, 1991
Spain	August 1, 1992	2 (13–15%)	Exogenous. Same rationale as previous case.	July 13, 1992

Table 2 (*continued*)

Country (1)	Date of implementation of VAT change (2)	Δ VAT rate (in percentage-points) (3)	Brief narrative (4)	Date of passage of VAT change (for exogenous cases only) (5)
Spain	January 1, 1995	1 (15–16%)	Exogenous. Deficits increased from 2.8–7.5% of GDP between 1989 and 1993 due to the recession of the early 1990s and a lax expenditures policy. In 1994 the Spanish government introduced several fiscal reforms to reduce the deficit.	December 31, 1994
Sweden	January 1, 1983	1.95 (21.51–23.46%)	Endogenous. The new government that took office in 1982 pursued a program aimed at increasing production and raising full employment (OECD, 1984). As part of this program, the VAT was increased to 23.46% in 1983 to cover the cost of restoring benefits cut by the previous government.	

Note: See Online Appendix for full tax narratives.

controls), they are classified as endogenous. In contrast, when changes in VAT rates during episodes of fiscal consolidation were motivated by inherited fiscal deficits and/or medium and long term solvency problems, they are classified as exogenous.

Table 2 summarizes the results of our analysis for the 19 fiscal consolidation tax rate changes.¹⁰ The table indicates country name (Column 1), date of implementation of the tax change (Column 2), change (in percentage points) of the VAT rate (Column 3), a brief narrative and a binary classification into endogenous and exogenous (Column 4). Lastly, column 5 shows (only for exogenous fiscal consolidation VAT rate changes) the date of passage of the law mandating the change in the VAT rate, which will be instrumental in evaluating (in Section 3.4) the role of expectations in the computation of tax multipliers.

Based on our narrative analysis, Table 2 identifies 75% (14 out of 19) of the fiscal consolidation VAT rate changes as exogenous to short-run output fluctuations. Most of the endogenous fiscal consolidation VAT rate changes (4 out of 5) correspond to cases where the VAT rate hike was part of a fiscal consolidation package that was also intended to stimulate economic activity by reducing other taxes, such as social security contributions. In one case (France in 2000), the VAT rate reduction was triggered by the idea that low deficits made it possible to lower existing (and high) tax rates. Section 3 will use these 14 exogenous tax rate changes to compute tax multipliers.

2.3. Cyclically adjusted revenues

Fig. 2 plots both the VAT rate and VAT cyclically adjusted revenue changes for each of our 14 countries.¹¹ The changes in VAT cyclically adjusted revenues have been computed using expression (3), quarterly VAT revenue, quarterly output growth data, and estimated values of η for indirect taxation from van den Noord (2000).¹²

Three features are worth noting. First, while VAT rate changes are discrete in nature and zero most of the time, VAT cyclically adjusted revenue changes show, by construction, a continuous-variable pattern.

Second, the contemporaneous correlation between changes in cyclically adjusted revenues and the tax rate is very low (0.04) and not significantly different from zero.¹³ Even when tax rate changes are not zero, the contemporaneous correlation between changes in cyclically adjusted revenues and the tax rate is rather low (0.21) and not significantly different from zero. If the comparison is restricted to exogenous fiscal consolidation tax rate changes, the correlation is -0.05 and not significantly different from zero.

Third, the analysis is complemented by looking at some basic dynamic properties of the cyclically adjusted measure, which helps us in understanding some inherent weaknesses of this revenue-based measure. Specifically, Fig. 2 reports two statistics for each country (see bottom note in each panel). The first is the Ljung–Box test statistic, which tests the null hypothesis that the data are independently distributed. The second is the autocorrelation coefficient (β) that results from estimating a simple AR1 process (i.e., $\Delta \text{cyclically-adjusted}_t = \alpha + \beta \cdot \Delta \text{cyclically-adjusted}_{t-1}$). The first statistic tells us that in almost 80% of the countries (11 out of 14) the changes in cyclically adjusted tax revenues are not independently distributed over time. Moreover, the second statistic tells us that, in more than 70% of the countries (10 out of 14), cyclically adjusted revenue changes are negatively autocorrelated at the quarterly frequency. This would suggest (under the common presumption that this measure is a proxy for changes in discretionary tax policy) that policymakers' discretionary policy changes are reversed every quarter! Since this is clearly highly implausible, it calls into question the usefulness of cyclically adjusted revenue changes as a proxy for changes in tax policy.

Section 4 will use the full sample of 42 VAT rate changes and the cyclically adjusted measures in the context of the Blanchard–Perotti identification to argue that, regardless of the identification strategy, cyclically adjusted revenues are a bad proxy for tax policy.

3. The effect of exogenous fiscal consolidation tax rate changes on output

This section studies the effect of exogenous fiscal consolidation VAT rate changes on economic activity. Our two benchmark specifications indicate that tax increases have a strong negative effect on output. We then conduct a myriad of robustness tests – alternative error structures, adding time dummies, changing the timing assumption, controlling for other fiscal variables, and taking into account the role of expectations – and conclude that our main results continue to hold in all cases.

¹⁰ The detailed narrative underlying Table 2 can be found in Online Appendix A.2.

¹¹ Note that Fig. 2 has been plotted for the common sample period and hence may not coincide with the sample period in Fig. 1.

¹² See Online Appendix A.1 for details.

¹³ One possible source of low contemporaneous correlation may be differences in timing. Suppose, for example, that revenues are turned over to the government with a one-quarter lag. Then there might be a low correlation between the contemporaneous changes in cyclically adjusted revenues and the tax rate, but a high correlation between the change in cyclically adjusted revenues and the lagged change in the tax rate. This conjecture, however, is not warranted as the correlation between cyclically adjusted revenues and lagged tax rates changes (up to four quarters) ranges between 0.07 and 0.0005.

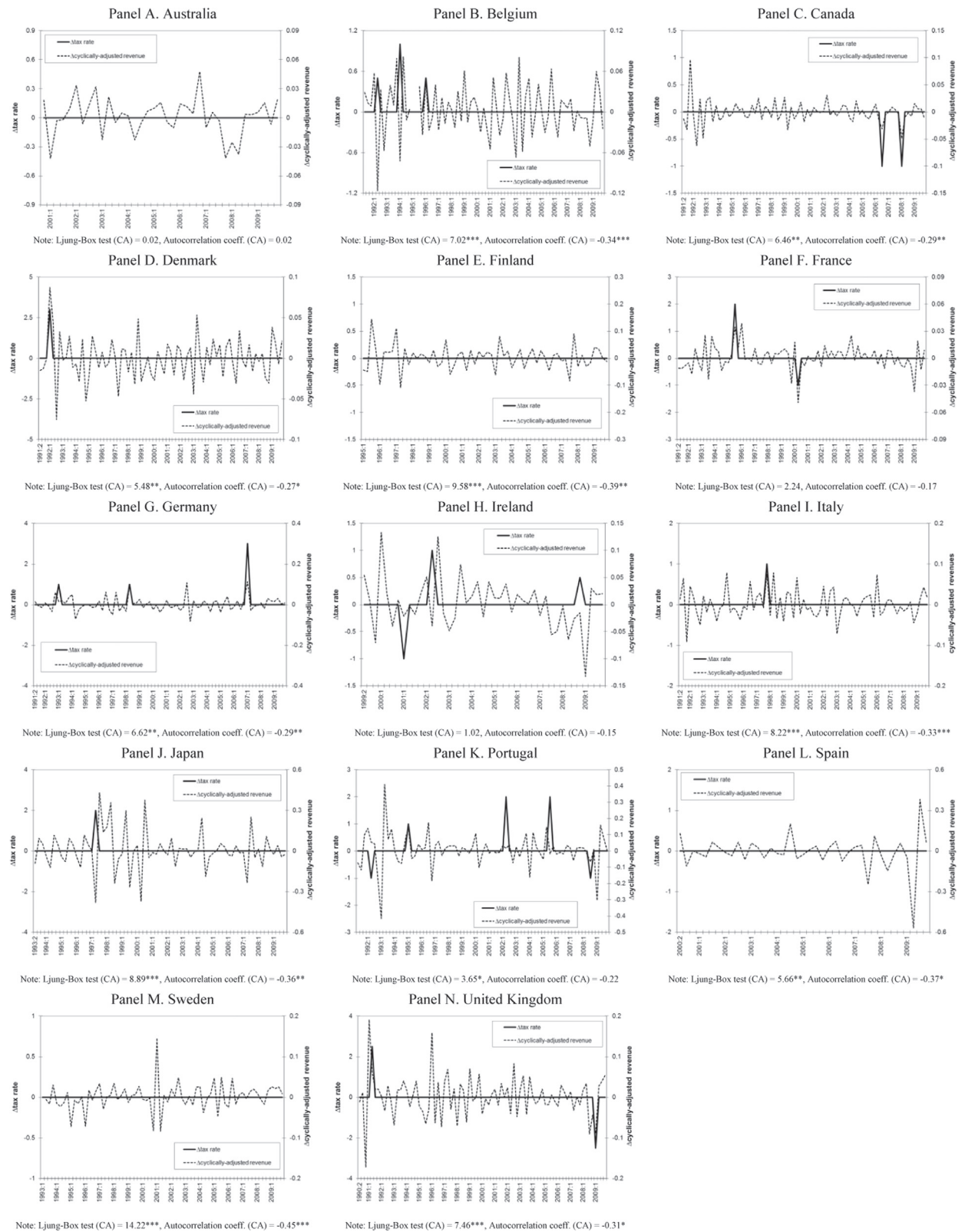


Fig. 2. Tax rate changes vs. cyclically adjusted revenue changes (CA). Note: *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

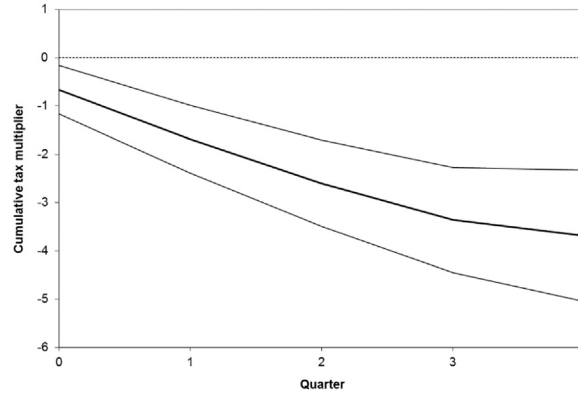


Fig. 3. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation, no controls. Notes: Country fixed effect panel regression. Standard errors are bootstrapped, heteroscedasticity-consistent (Huber-White), and clustered by country. Observations: 1304.

3.1. Benchmark specifications

The analysis begins with the estimation of the effect of tax rate changes on economic activity using, as in RR, the following basic specification:

$$\Delta y_{i,t} = \alpha + \sum_{j=0}^J \beta_j \Delta t_{i,t-j} + \sum_{i=1}^{I-1} \beta_i d_i + \mu_{i,t}, \quad (4)$$

where $J=4$, $I=14$, y is the logarithm of real GDP (and Δy is thus the real GDP growth rate, expressed as the difference in logarithms), Δt is the change in tax rate in percentage point terms and represents exogenous fiscal consolidation tax rate changes, α is a constant term, d is a country dummy, and μ is the error term. In estimating Eq. (4), four lags are included (i.e., $J=4$).¹⁴

This approach should yield unbiased reduced-form impact estimates of tax rate changes on output (see top-left cell in Table 1) given (i) the exogenous nature of tax rate changes and (ii) that the tax variable used is a tax instrument (as opposed to a revenue-based one, such as the cyclically adjusted revenue).¹⁵

While conceptually appropriate, a drawback of using changes in tax rates as the independent variable in Eq. (4) is that the estimated coefficients β do not correspond to the usual tax multiplier for GDP (i.e., the response of GDP to a unit increase in tax revenues). In other words, the coefficients β link the change in GDP to the change in the tax rate and not in tax revenue. To think in terms of standard multipliers, the multiplier for time horizon J is defined as¹⁶

$$\text{Tax multiplier } (J) \equiv \frac{\sum_{j=0}^J \Delta Y_j}{\sum_{j=0}^J \Delta R_j}. \quad (5)$$

The tax multiplier defined above is computed by exploiting the typical relationship of tax revenues to the tax rate. Let R be real VAT revenue, Y real output, I “implicit” VAT rate (defined as R/Y), and e the average relationship between I and t (i.e., $e \equiv I/t$). Following Barro and Redlick (2011, pp. 80–81), the tax multiplier at different time horizons J is then computed as¹⁷

$$\text{Tax multiplier } (J) = \frac{\omega_J}{e + \omega_J \cdot \bar{I}_J}, \quad (6)$$

where ω_J represents the cumulative effect on the growth rate of output of a one percentage-point shock in taxes. In the case of specification (4), $\omega_J = \sum_{j=0}^J \beta_j$. $\bar{I}_J$ measures the average “implicit” VAT rate (i.e., $\bar{I}_J \equiv (\sum_{j=0}^J I_j)/J$).¹⁸

¹⁴ The selection of four lags balances the need to account for a sufficiently long structure of lags, in order to study the effect of fiscal consolidations, and the need to preserve most of our tax rate changes. Unfortunately, longer lag structures force us to drop data points from an already small set of tax rate changes. Having said that, the results for the case of eight and twelve lags remain almost the same as in the four quarter specification. The main difference is that the cumulative multiplier becomes more negative starting in the eight quarter. Figures showing the multipliers for the eight and twelve lag estimations are not shown for the sake of brevity.

¹⁵ Given concerns about the non-normal distribution of tax rate changes (which are zero most of the time) and small sample considerations, standard errors are computed using bootstrap techniques. In particular, standard errors are calculated from the average of 10,000 draws of the coefficient vector from a multivariate normal distribution with mean and variance–covariance matrix equal to the point estimates and variance–covariance matrix of the regression coefficients.

¹⁶ No substantial difference in our results would arise if the median real deposit rate were used for discounting purposes.

¹⁷ See Online Appendix A.3 for the derivation of the next two expressions.

¹⁸ We use sample average value of e (38%) and the “implicit” VAT rate (6%) for our tax multiplier calculations. While there is some variation across countries, the variability is fairly small. The standard deviations of e and the “implicit” VAT rate are 6% and 2%, respectively.

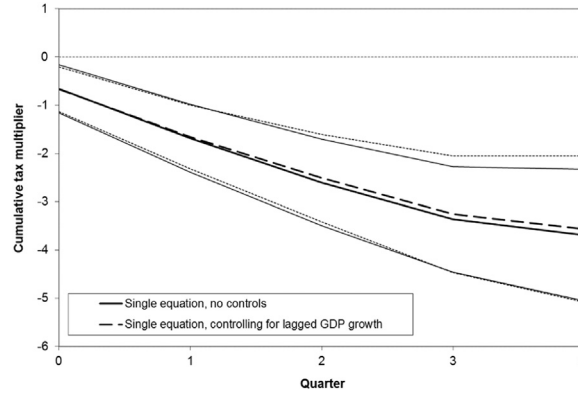


Fig. 4. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation, controls vs. no controls. Notes: Country fixed effect panel regression. Standard errors are bootstrapped, heteroscedasticity-consistent (Huber-White), and clustered by country. Observations: 1304 (no controls), 1288 (controlling for lagged GDP growth).

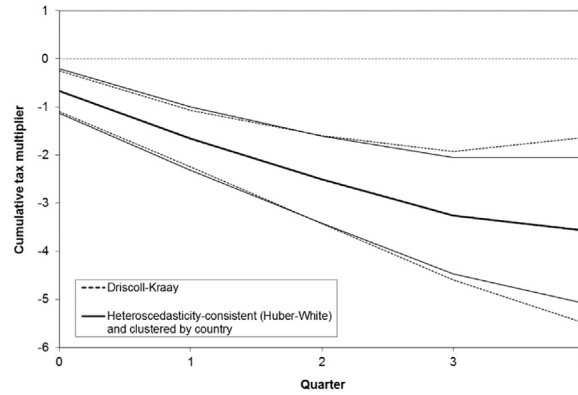


Fig. 5. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation with controls, alternative structure of errors. Notes: Country fixed effect panel regression. Standard errors are (i) solid lines: bootstrapped, heteroscedasticity-consistent (Huber-White), and clustered by country, and (ii) dashed lines: bootstrapped, cross-sectional (spatial) and temporal dependence robust (Driscoll-Kraay). Observations: 1288.

Using a first-order approximation of (6), the standard error of the tax multiplier can be written as

$$\text{Tax multiplier}_{SE}(J) = \frac{e}{(e + \omega_J \cdot \bar{I}_J)} \omega_J^{SE}, \quad (7)$$

where ω_J^{SE} is the standard error of ω_J .

Fig. 3 uses estimates from (4) to illustrate the tax multiplier together with one-standard-error bands. The multiplier is consistently and significantly negative. In particular, the multiplier is -0.66 on impact and increases (in absolute value) with longer horizons until reaching -3.69 after four quarters.¹⁹

The specification (4) is now enriched by adding four lags of output growth:

$$\Delta y_{i,t} = \alpha + \sum_{j=0}^J \beta_j \Delta t_{i,t-j} + \sum_{n=1}^N \beta_n \Delta y_{i,t-n} + \sum_{i=1}^{I-1} \beta_i d_i + \mu_{i,t}. \quad (8)$$

where $J=4$ and $N=4$. As discussed in RR, including lagged growth helps in controlling for the normal dynamics of output as well as providing a simple way of controlling for other factors affecting output, most likely serially correlated. Using the estimates from (8), Fig. 4 shows the corresponding tax multiplier.^{20,21} For comparison purposes, this figure also reports the findings from the specification without GDP growth lags (i.e., specification (4)). Results are virtually unchanged both in terms of point estimates and precision.

¹⁹ Results are virtually not affected if country dummies are not included. Results are not shown for brevity.

²⁰ As in the case of regression (4), standard errors in specification (8) are computed using bootstrap techniques.

²¹ The impulse response function of a shock in taxes in specification (8) includes both the direct impact of taxes on output (captured by the coefficients β_j) and the indirect cumulative effect through the lags of output growth (captured by the coefficients β_n).

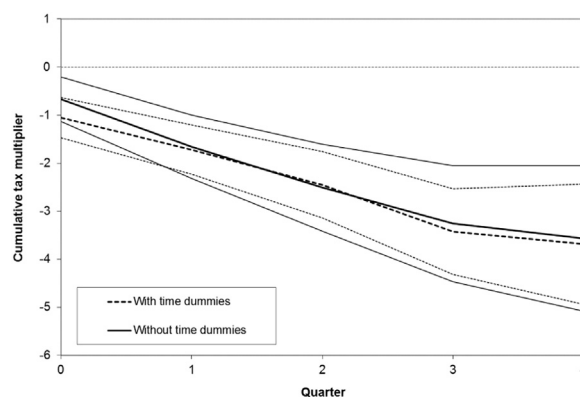


Fig. 6. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation with controls, with and without time dummies. Notes: Country fixed effect panel regression. Standard errors are bootstrapped, heteroscedasticity-consistent (Huber-White), and clustered by country. Observations: 1288.

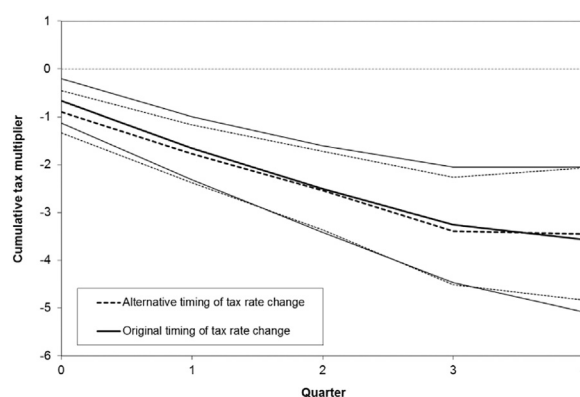


Fig. 7. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation with controls, alternative timing of tax rate change. Notes: Country fixed effect panel regression. Standard errors are bootstrapped, heteroscedasticity-consistent (Huber-White), and clustered by country. Observations: 1288.

3.2. Robustness tests

Several robustness tests are now performed, which build upon the specification in equation (8).²²

3.2.1. Alternative error structure

To account for an alternative error structure, Fig. 5 shows the results when using Driscoll and Kraay (1998) standard errors, which allow the error term to be serially and cross-sectionally correlated. The results are not affected.

3.2.2. Time dummies

Time dummies are now introduced. Fig. 6 shows that, relative to our benchmark specification with no time dummies, the impact multiplier is marginally larger (in absolute value) on impact, but remains virtually unchanged afterwards.

3.2.3. Timing

As is standard in the literature, our benchmark specification – given by (8) – imputes the tax rate change to the quarter in which it was effectively implemented (independently of whether such a tax rate change was implemented at the beginning, middle, or end of the quarter). One may argue, however, that if the tax rate change was, for example, implemented towards the end of the quarter, the impact on output should only be observed in the following quarter. If so, this would bias the multiplier downwards (in absolute value).

Table 2 shows that while more than 70% (or 10 out of 14) of exogenous fiscal consolidation tax rate changes were implemented in the very first day of each quarter, more than 20% (or 3 out of 14) were implemented in the last month of each quarter.²³ Only one tax rate change was implemented in the second month of the quarter. To account for these

²² Similar results are obtained if the first benchmark specification, given by (4), is used. Results are not shown for the sake of brevity.

²³ In six episodes, the change in the VAT was implemented on January 1, reflecting the start of the new calendar year.

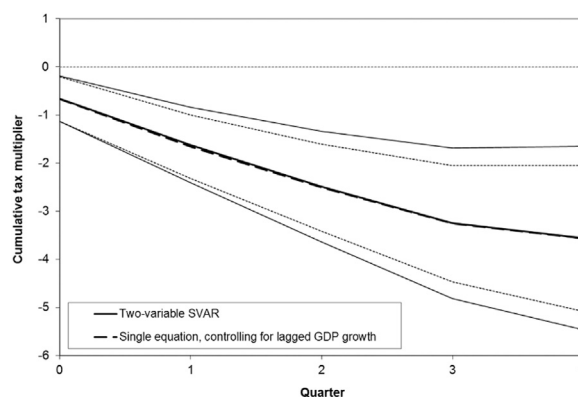


Fig. 8. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation with controls vs. two-variable SVAR. Notes: The two variables in SVAR are exogenous fiscal consolidation tax rate change and GDP growth. We assume no contemporaneous effect of output on tax rate change. Observations: 1288 (controlling for lagged GDP growth) and 1284 (SVAR).

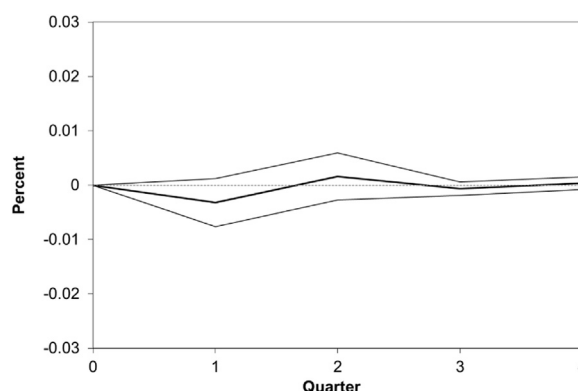


Fig. 9. Response of exogenous fiscal consolidation tax rate to GDP shock: 1% impulse response function, two-variable SVAR. Notes: The two variables in SVAR are exogenous fiscal consolidation tax rate change and GDP growth. We assume no contemporaneous effect of output on tax rate change. Observations: 1284.

instances, Fig. 7 shows the results of an alternative measure of tax rate changes in which tax rate changes implemented in the last month of a quarter are imputed to the following quarter.²⁴ The results remain virtually unchanged.

3.2.4. Two-variable SVAR

Lastly – and following RR – we use a natural variation of specification (8): a two-variable pooled SVAR with real GDP growth and exogenous fiscal consolidation tax rate changes. In keeping with the regression specifications on the contemporaneous relationship between tax rate and output, the tax rate changes are allowed to affect output contemporaneously but not vice versa. Of course, the construction of a narrative identifying exogenous fiscal consolidation tax rate change makes the use of an SVAR approach trivial. In other words, as long as the narrative is correct, one would not need to run a system of equations and make any assumptions about the contemporaneous relation between exogenous fiscal consolidation tax rate changes and real GDP growth rate in order to identify a fiscal shock.

Having said that, we follow RR and the literature more generally and still estimate an SVAR for several reasons: (i) to provide some empirical evidence that will “validate” our narrative approach for exogenous fiscal consolidation tax rate changes (i.e., analyze whether exogenous fiscal consolidation tax rate changes are, indeed, unrelated to recent past output fluctuations); (ii) to help address potential endogeneity concerns when including other controls variables, such as government spending; and (iii) to compare our exogenous tax measure with other measures of tax policy changes; in particular with the commonly used cyclically adjusted revenue changes.

The first item is now addressed (the second will be addressed in Section 3.3 and the third in Section 4). Fig. 8 shows the tax multiplier when using the two-variable SVAR.²⁵ The multiplier is virtually identical to the one obtained using specification (8).²⁶

²⁴ Virtually identical results are obtained if one also imputes the only tax rate change that was implemented in the second month of a quarter to the following quarter.

²⁵ As in (4) and (8), four lags are used in the estimation of the two-variable SVAR.

²⁶ For the sake of comparison with Fig. 8, the plot is restricted to four quarters (since in the case with the single equation approach the cumulative multiplier beyond period 4 would be constant).

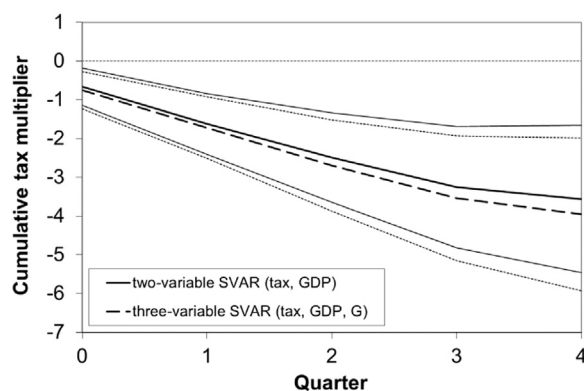


Fig. 10. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, two-variable vs. three-variable SVAR. Notes: The two-variable SVAR includes exogenous fiscal consolidation tax rate changes and GDP growth. The three-variable SVAR includes exogenous fiscal consolidation tax rate changes, fiscal consolidation government expenditure growth, and GDP growth. We assume no contemporaneous effect of output on fiscal variables. We also assume no contemporaneous correlation between tax rate changes and changes in government expenditure. Observations: 1284 (two-variable SVAR), 1063 (three-variable SVAR).

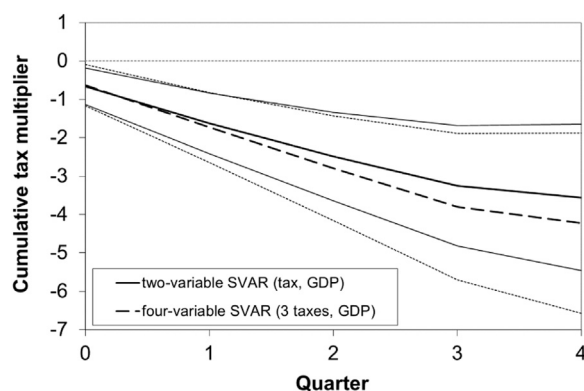


Fig. 11. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, two-variable vs. four-variable SVAR. Notes: The two-variable SVAR includes exogenous fiscal consolidation VAT rate changes and GDP growth. The four-variable SVAR includes exogenous fiscal consolidation tax rates changes (value-added, top individual income, and top corporate income) and GDP growth. Observations: 1284 (two-variable SVAR), 975 (four-variable SVAR).

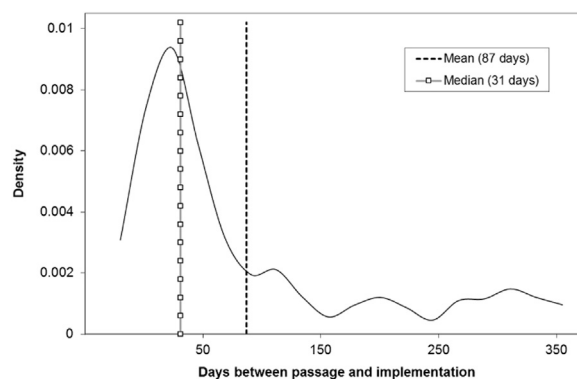


Fig. 12. Density function of days between passage and implementation of exogenous fiscal consolidation tax rate shock.

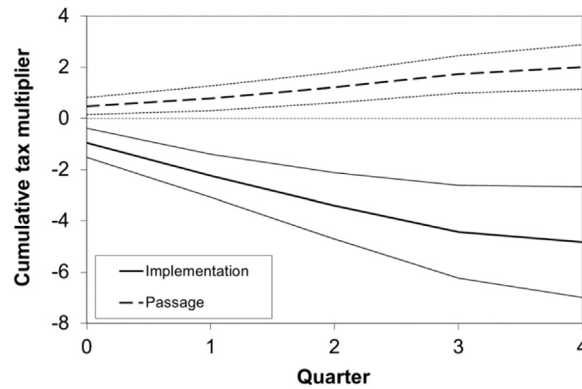


Fig. 13. Cumulative tax multiplier: Exogenous fiscal consolidation tax rate shock, single equation with controls, controlling for the announcement date. Notes: Country fixed effect panel regression. Standard errors are bootstrapped, heteroscedasticity-consistent (Huber-White), and clustered by country. Observations: 1288.

Using all tax rate changes

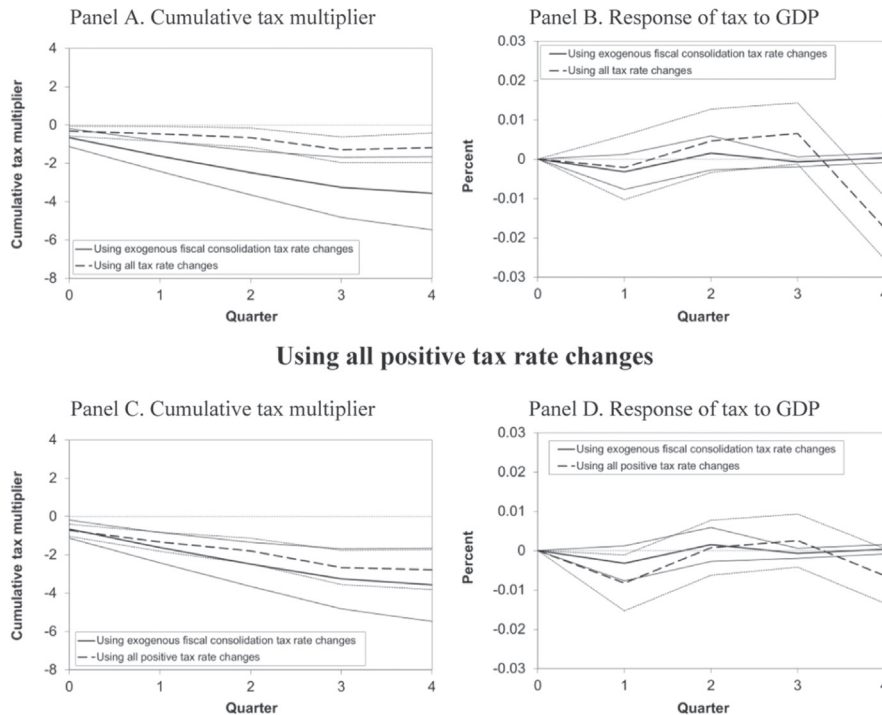


Fig. 14. Cumulative tax multiplier and 1% impulse response functions, two-variable SVAR, all and positive tax rate changes vs. exogenous fiscal consolidation tax rate changes. Notes: Panels A and C show the cumulative tax multiplier and panels B and D the 1% impulse response of tax rate to a GDP shock. We assume no contemporaneous effect of output on fiscal variables. Observations: 1284.

To empirically “validate” our narrative approach, Fig. 9 shows the impulse response function of the tax series to a shock in GDP. After a 1% innovation in output, tax rates barely fluctuate between 0.002 and -0.003 . The effect is highly statistically insignificant at all time horizons. The p -value for the test of the null hypothesis that output does not Granger-cause the tax series is 0.96. This finding strongly supports our conjecture that tax rate changes identified using exogenous fiscal consolidation episodes (i.e., through historical narratives) are, indeed, unrelated to past output fluctuations.

3.3. Controlling for changes in other fiscal variables

Following RR, we now control for changes in other fiscal variables such as government spending and other taxes. Government spending, of course, may affect output on its own and hence bias the tax multiplier. For instance, consider an episode of fiscal consolidation where government spending typically falls at the same time that tax rates increase. Omitting

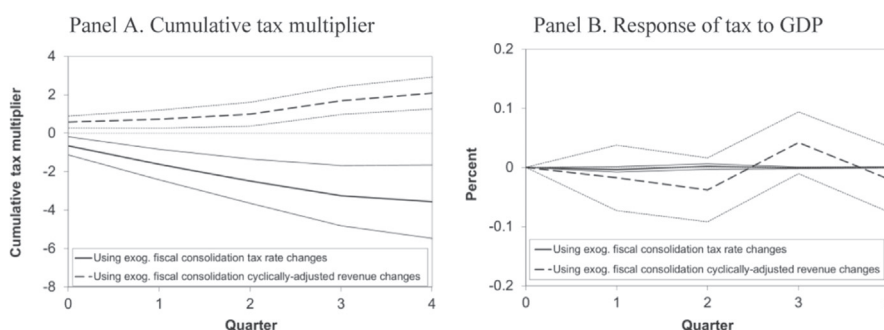


Fig. 15. Cumulative tax multiplier and 1% impulse response functions, two-variable SVAR, exogenous fiscal consolidation cyclically adjusted revenue changes vs. exogenous fiscal consolidation tax rate changes. Notes: Panel A shows the cumulative tax multiplier and panel B the 1% impulse response of tax rate to a GDP shock to a GDP shock. We assume no contemporaneous effect of output on fiscal variables. Observations: 856.

the former would bias the tax multiplier downwards, giving the false idea that tax rate hikes are more contractionary than they actually are. The opposite would be true when government spending increases together with tax rate hikes, perhaps in an attempt to offset the potential contractionary effect of tax rate hikes on output. Since the data do not show any systematic contemporaneous relation (at a quarterly frequency) between changes in government spending and tax rates during episodes of fiscal consolidation (the correlation is -0.02 and not statistically different from zero), this issue should not be a problem, but a formal analysis is clearly warranted.

In order to control for government spending, we use a three-variable SVAR specification with four lags of real GDP growth rate, exogenous fiscal consolidation tax rate changes, and government spending changes occurring during periods of fiscal consolidation. In keeping with the regression specifications on the contemporaneous relationship between tax rate and output, tax rate changes are allowed to affect output contemporaneously but not viceversa. The same is true for government spending (i.e., government spending can affect output contemporaneously but not viceversa). As is common practice, no contemporaneous relation is assumed between government spending and tax rates.²⁷

Fig. 10 shows the results of the three-variable SVAR. The tax multiplier remains virtually unchanged after controlling for government spending. This was to be expected to the extent that, as mentioned above, the contemporaneous correlation is essentially zero. While there could still be some dynamic feedback from changes in tax rates to government spending, Fig. 10 suggests that this not the case either.²⁸

With respect to changes in other taxes, we now control for corporate and individual income tax rates. Given the lack of readily available data on average marginal income and corporate tax rates for the 14 countries used in this study, top marginal corporate and individual income tax rates are used. To this effect, the benchmark two-variable SVAR is extended to a four-variable SVAR specification with the real GDP growth rate and exogenous fiscal consolidation tax rate changes (value-added, corporate, and income).²⁹ Fig. 11 shows the tax multiplier with and without these additional tax rates. The impact effect remains virtually unchanged and the cumulative multiplier is also virtually the same.

In sum, the benchmark results remain essentially unchanged after controlling for other relevant fiscal variables.

3.4. The role of expectations

In deriving our measure of tax multipliers, the date of tax changes corresponds to the date in which they were actually implemented. Expectations, however, could also matter. As discussed in RR and Alesina et al. (2015), announcements may lead to anticipated effects of tax changes.

To explore this issue, Fig. 12 shows the density function for the number of days between passage and implementation of exogenous fiscal consolidation tax rate changes. The lag between passage and implementation is relatively short, with a median and average lag of about one and three months, respectively.

²⁷ As in Blanchard and Perotti (2002) and Favero and Giavazzi (2007), results remain virtually unchanged when α_{gt} (contemporaneous effect of government spending on tax rates) and α_{tg} (contemporaneous effect of tax rates on government spending) are estimated using, for example, two dynamic least squares dummy variable regressions with four lags of tax rates and four lags of government spending in each regression and including these estimates, one at a time, in the three-variable SVAR analysis ($\alpha_{gt} = 0.01$ ($t = 1.06$) and $\alpha_{tg} = 0.21$ ($t = 1.10$)). Results are not shown for the sake of brevity.

²⁸ If all changes in government spending were included instead of only those associated with fiscal consolidation episodes, the tax multiplier would remain virtually unchanged. These figures are not reported for the sake of brevity.

²⁹ As before, tax rate changes are allowed to affect output contemporaneously but not the other way around. Four lags of all variables are included. It is also assumed that the contemporaneous relations between the different tax rate changes in the matrix of endogenous variables are zero. The results barely change when using estimates from a least squares dummy variable model. These plots are not shown for brevity's sake.

In order to control for possible anticipated effects arising from announcements, we follow RR's strategy of adding to our previous specification, given by Eq. (8), a “news” term and its corresponding lags. Formally,

$$\Delta y_{i,t} = \alpha + \sum_{j=0}^J \beta_j \Delta t_{i,t-j} + \sum_{m=0}^J \beta_m \Delta \text{News}_{i,t-m} + \sum_{n=1}^N \beta_n \Delta y_{i,t-n} + \sum_{i=1}^{I-1} \beta_i d_i + \mu_{i,t}, \quad (9)$$

where $J=4$, $N=4$, $I=14$, and the variable “News” represents the expected change of the tax rate at the time of the official passage of the law. As in the case of the actual tax change, four lags are included for the “News” term. Fig. 13 then shows the response of output to the passage of the law and to the implementation of the law (i.e., the tax multiplier that has been reported all along).

Interestingly, Fig. 13 shows that, at the time of the passing of the law, output increases. Most importantly for our purposes, the contractionary effect of tax increases in response to the implementation of the law remains valid even after fiscal news are controlled for.

In conclusion, news about tax changes (proxied by the passage of tax laws) seem to lead to intertemporal substitution effects in the sense that output increases in response to the news even though it falls when the law is actually implemented. The estimates of the tax multiplier itself, however, remain robust.

4. Comparison with alternative empirical strategies

This section examines the implications of using alternative empirical strategies for the identification of exogenous fiscal shocks and tax policy measurement. In terms of Table 1, notice that the analysis above has dealt exclusively with the top-left cell. This section deals with the cases captured by the other three cells. For comparison purposes, the plots in this section also report the findings from the two-variable SVAR specification with exogenous fiscal consolidation tax rate changes illustrated in Fig. 8.³⁰

4.1. Tax rate and Blanchard–Perotti

Consider first the implications of using the VAT rate, a tax instrument, and the Blanchard–Perotti identification strategy (bottom-left cell in Table 1). In other words, we use all tax rate changes – not only those associated with exogenous fiscal consolidation episodes – and follow the structure proposed by Blanchard and Perotti (2002) to identify fiscal shocks.³¹ Fig. 14, panels A and B, displays the results. Panel A shows that the time profile of the multiplier is quite similar to our benchmark, but considerably smaller (in absolute value). In particular, the impact effect, while still contractionary, is halved (i.e., falls in absolute value from 0.66 to 0.31) and is borderline significant. The maximum contractionary effect is reduced to about a third, from 3.56 to 1.29 (in absolute value). Panel B indicates that tax rate changes tend to respond significantly to output changes, especially after four quarters. The p -value for the test of the null hypothesis that a positive shock in output does not reduce the tax rate in the fourth quarter is 0.015.

One could argue that the difference in tax multipliers in Fig. 14, Panel A, may reflect the asymmetric impact of tax rate changes depending on whether they are tax cuts or hikes. Since all of our exogenous fiscal consolidation tax rate changes are tax hikes, we now look at the case in which *all positive* tax rate changes are used and follow the Blanchard–Perotti identification scheme.³² Panels C and D in Fig. 14 display the results. Panel C confirms the findings in Panel A; that is, the time profile of the multiplier is similar to our benchmark but its size (in absolute value) is much smaller.

4.2. Cyclically adjusted revenues and narrative

Consider now the case captured by the top-right cell in Table 1. In other words, VAT cyclically adjusted revenue changes are used as the tax instrument and the exogenous fiscal consolidation tax rate changes as the identification strategy.³³ Since cyclically adjusted revenue changes are included only for those quarters with exogenous fiscal consolidation tax rate changes, the exercise truly compares alternative tax policy measures.³⁴ Panels A and B in Fig. 15 display the results. Panel A shows a profile for the tax multiplier very different from the one obtained so far. In particular, the impact multiplier is very small (borderline zero) and expansionary after two quarters.³⁵ Interestingly, and in line with our previous results, Panel B

³⁰ The SVAR specification is used as the benchmark in order to have a valid comparison because all of the results in this section are derived in the context of an SVAR.

³¹ Recall that our VAT rate sample includes 42 tax rate changes.

³² Recall that our sample includes 32 tax rate increases.

³³ See Online Appendix A.3 for the computation of the multiplier and its standard deviation.

³⁴ Similar results are obtained including cyclically adjusted revenue changes for the year (as opposed to just the quarter) with exogenous fiscal consolidation tax rate changes. In this case, the tax multiplier is much smaller (in absolute value). Results are not shown for the sake of brevity.

³⁵ The impact effect is in line with the results in Perotti (2004), Favero and Giavazzi (2012), Caldara (2011), and Ilzetzi (2011), who use the same instrument but a different methodology (SVAR). In contrast, Mertens and Ravn (2013) find large multipliers using narratively identified tax changes as proxies for structural tax shocks in a SVAR set-up.

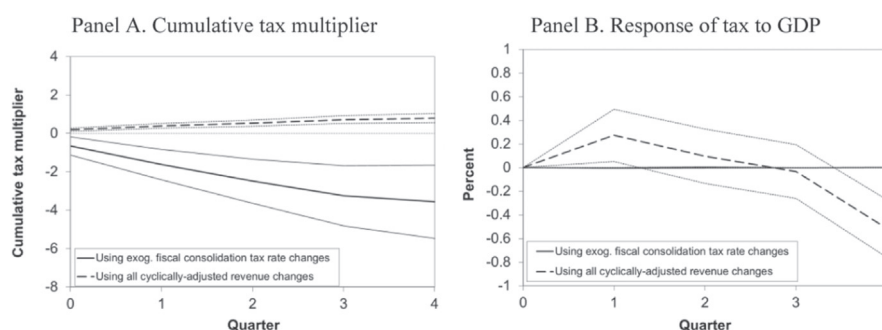


Fig. 16. Cumulative tax multiplier and 1% impulse response functions, two-variable SVAR, cyclically adjusted revenue changes vs. exogenous fiscal consolidation tax rate changes. Notes: Panel A shows the cumulative tax multiplier and panel B the 1% impulse response of tax rate to a GDP shock. We assume no contemporaneous effect of output on fiscal variables. Observations: 856.

seems to empirically “validate” the use of exogenous fiscal consolidation as a means of identifying exogenous fiscal shocks. The p -value for the test of the null hypothesis that output does not granger-cause changes in tax rates is 0.75.

4.3. Cyclically adjusted revenues and Blanchard–Perotti

Lastly, consider the case of VAT cyclically adjusted revenue changes and the Blanchard–Perotti identification strategy (bottom-right cell in Table 1). As discussed in the Introduction, this strategy is by far the most commonly used in the literature. Fig. 16, panels A and B, display the results. Panel A provides evidence that (as in panel A in Fig. 15) the impact effect on output is not only non-contractionary, but, in fact, expansionary! A unit shock increase in VAT revenue collection increases output by 0.19 on impact and about 0.80 after a year.

Furthermore, when using this revenue-based measure together with the Blanchard–Perotti identification strategy, our results indicate that cyclically adjusted revenue changes tend to respond to output changes (Panel B). This casts severe doubts on the validity of Blanchard–Perotti as a tool for identifying fiscal shocks.

4.4. Implications of our findings

This section's findings clearly support our concerns regarding the use of both cyclically adjusted revenue measures to determine changes in tax policy as well as the use of econometric structural assumptions to identify exogenous fiscal shocks.

In terms of *identification*, our empirical evidence favors the use of “natural experiments” as a more accurate way of assessing fiscal policy changes not resulting from policymakers' response to output fluctuations (see Fig. 14, panel B for tax rates and panel B in Figs. 15 and 16 for cyclically adjusted revenues).

Regarding the (much less explored) issue of *measurement* of tax policy, we conclude (complementing our findings from Section 2.3) that there is really no good substitute for data on tax rates. Our main point – the importance of using tax rates to measure tax policy – is, in fact, independent of the particular identification strategy, as it applies equally to the analysis of fiscal shocks identified through narratives à la RR and SVAR identification à la Blanchard–Perotti. The critical importance of this issue is reflected in the very different tax multipliers that result. Tax rate increases are always contractionary (Fig. 14, Panels A and C). Tax multipliers based on cyclically adjusted revenues, however, are either very small or actually positive (Panel A in Figs. 15 and 16), suggesting that tax increases actually raise output. In other words, not using tax rates may lead to a very misleading value of the tax multiplier.

5. Conclusions

This paper contributes to the empirical literature on tax multipliers. Specifically, the paper first revisits the issue of the identification of exogenous fiscal policy shocks (i.e., fiscal policy changes that are not the result of policymakers responding to output fluctuations). The findings support Romer and Romer's (2010) narrative approach as a more convincing strategy than SVAR to truly identify exogenous fiscal shocks. The paper then explores the (rarely addressed) implications of tax policy measurement (i.e., obtaining a tax policy variable that is under the direct control of policymakers). For this purpose, the paper introduces a novel value-added tax rate dataset and the associated cyclically adjusted revenues at a quarterly frequency for 14 industrial countries for the period 1980–2009. The results strongly support the use of tax rates as a true measure of tax policy as opposed to widely used revenue-based measures, such as cyclically adjusted revenues. The conclusion is that the latter is a poor proxy for changes in tax rates since it shows a very erratic pattern (on many occasions even implying that the presumably discretionary policy changes are reversed every quarter!). The paper's main point – the importance of using tax rates as a proxy for tax policy – is, however, independent of the particular identification strategy, as it applies equally to the analysis of fiscal shocks identified through narratives à la Romer–Romer as well as to SVAR

identification. Tax rate increases are always contractionary. However, tax multipliers based on cyclically adjusted revenues are, as suggested by some papers in this area, close to zero or even positive.

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Appendix A. Supplementary data

Supplementary data associated with this paper can be found in the online version at <http://dx.doi.org/10.1016/j.jmoneco.2016.03.003>.

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